

#1

(a)

$$P(A) = 1-p$$

$$P(B) = p\delta_0 + (1-p)\delta_1$$

$$P(A \cap B) = (1-p)\delta_1$$

$$P(A)P(B) = (1-p)p\delta_0 + (1-p)^2\delta_1$$

if A, B independent, then $P(A)P(B) = P(A \cap B)$

$$\Rightarrow (1-p)p\delta_0 + (1-p)^2\delta_1 - (1-p)\delta_1 = 0$$

$$\Rightarrow (1-p)[p\delta_0 + (1-p)\delta_1 - \delta_1] = 0$$

$$\Rightarrow p(1-p)(\delta_0 - \delta_1) = 0 \Rightarrow \text{when } p=0 \vee p=1 \vee \delta_0 = \delta_1, A, B \text{ independent}$$

(b)

$$P(A|C) = \frac{(1-p)(a_1 + \varepsilon_1)}{p(a_0 + \varepsilon_0) + (1-p)(a_1 + \varepsilon_1)}$$

$$P(B|C) = \frac{p\varepsilon_0 + (1-p)\varepsilon_1}{p(a_0 + \varepsilon_0) + (1-p)(a_1 + \varepsilon_1)}$$

$$P(A \cap B|C) = \frac{(1-p)\varepsilon_1}{p(a_0 + \varepsilon_0) + (1-p)(a_1 + \varepsilon_1)}$$

$$P(A|C)P(B|C) = \frac{(1-p)(a_1 + \varepsilon_1)[p\varepsilon_0 + (1-p)\varepsilon_1]}{[p(a_0 + \varepsilon_0) + (1-p)(a_1 + \varepsilon_1)]^2}$$

if A, B independent given the event C , then

$$P(A \cap B|C) = P(A|C)P(B|C)$$

$$\Rightarrow \frac{(1-p)(a_1 + \varepsilon_1)[p\varepsilon_0 + (1-p)\varepsilon_1]}{[p(a_0 + \varepsilon_0) + (1-p)(a_1 + \varepsilon_1)]^2} = \frac{(1-p)\varepsilon_1}{p(a_0 + \varepsilon_0) + (1-p)(a_1 + \varepsilon_1)}$$

$$\Rightarrow (1-p)(a_1 + \varepsilon_1)[p\varepsilon_0 + (1-p)\varepsilon_1] - (1-p)\varepsilon_1[p(a_0 + \varepsilon_0) + (1-p)(a_1 + \varepsilon_1)]$$

$$\Rightarrow (1-p)p(a_1\varepsilon_0 - a_0\varepsilon_1) = 0 \Rightarrow \text{That is, when } p=1 \vee p=0 \vee a_1\varepsilon_0 = a_0\varepsilon_1$$

A, B independent

(c)

$$P(\text{correct}) = P(\text{correct}|0)P(0) + P(\text{correct}|1)P(1)$$

$$= P(1 - \varepsilon_0 - a_0) + (1-p)(1 - \varepsilon_1 - a_1)$$

2

(a)

$$P(X=n) = \sum_{m=0}^{\infty} P(X=n | Z=n+m) P(Z=n+m)$$

$$\begin{cases} P(X=n | Z=n+m) = \binom{n+m}{n} p^n (1-p)^m \\ P(Z=n+m) = \frac{(\lambda T)^{n+m} e^{-\lambda T}}{(n+m)!} \end{cases}$$

$$\begin{aligned} \Rightarrow P(X=n) &= \sum_{m=0}^{\infty} \binom{n+m}{n} p^n (1-p)^m \frac{(\lambda T)^{n+m} e^{-\lambda T}}{(n+m)!} \\ &= p^n \frac{(\lambda T)^n e^{-\lambda T}}{n!} \sum_{m=0}^{\infty} \frac{((1-p)\lambda T)^m}{m!} \\ &= p^n \frac{(\lambda T)^n e^{-\lambda T}}{n!} e^{(1-p)\lambda T} \\ &= \frac{(p\lambda T)^n e^{-p\lambda T}}{n!} \quad \# \end{aligned}$$

We show that the number of transmitted in that same time interval has a Poisson PMF parameter $p\lambda$.

(b)

$$P(X_i = k) = (1-p)^{k-1} p$$

$$P(X \leq k) = P(\max(X_1, X_2, \dots, X_n) \leq k)$$

$$\Rightarrow P(X \leq k) = P(X_1 \leq k) P(X_2 \leq k) \dots P(X_n \leq k)$$

$$\Rightarrow P(X \leq k) = [1 - (1-p)^k]^n$$

To find $P(X=k)$, we need to compute $P(X \leq k) - P(X \leq k-1)$

$$\Rightarrow \text{due to } P(X \leq k) = [1 - (1-p)^k]^n \Rightarrow P(X \leq k-1) = [1 - (1-p)^{k-1}]^n$$

$$\Rightarrow P(X=k) = [1 - (1-p)^k]^n - [1 - (1-p)^{k-1}]^n \quad \#$$

$$P(Y \geq k) = P(\min(X_1, X_2, \dots, X_n) \geq k)$$

$$P(Y \geq k) = P(X_1 \geq k) P(X_2 \geq k) \dots P(X_n \geq k) \text{ (due to independent)}$$

$$P(Y \geq k) = (1-p)^{kn}$$

$$P(Y=k) = P(Y \geq k) - P(Y \geq k+1) = (1-p)^{(k-1)n} - (1-p)^{kn} \neq$$

最早成功的 r.v 是 Y #

$$(c) \quad P(X = X_{s_2}) = \binom{123}{X_{s_2}} p_r^{X_{s_2}} (1-p_r)^{123-X_{s_2}} \neq$$

$$\begin{aligned} \#3_{(a)} \quad LHS &= H\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = -\left(\frac{1}{2} \ln \frac{1}{2} + \frac{1}{3} \ln \frac{1}{3} + \frac{1}{6} \ln \frac{1}{6}\right) \\ RHS &= H\left(\frac{1}{2}, \frac{1}{2}\right) + \frac{1}{2} H\left(\frac{2}{3}, \frac{1}{3}\right) = -\left(\frac{1}{2} \ln \frac{1}{2} + \frac{1}{2} \ln \frac{1}{2}\right) - \frac{1}{2} \left(\frac{2}{3} \ln \frac{2}{3} + \frac{1}{3} \ln \frac{1}{3}\right) \\ &= -\left(\ln \frac{1}{2} + \frac{1}{3} \ln \frac{2}{3} + \frac{1}{6} \ln \frac{1}{3}\right) \\ &= -\left(\ln \frac{1}{2} + \frac{1}{3} \ln \frac{1}{3} - \frac{1}{3} \ln \frac{1}{2} + \frac{1}{6} \ln \frac{1}{2} - \frac{1}{6} \ln \frac{1}{2} + \frac{1}{6} \ln \frac{1}{3}\right) \\ &= -\left(\frac{1}{2} \ln \frac{1}{2} + \frac{1}{3} \ln \frac{1}{3} + \frac{1}{6} \ln \frac{1}{6}\right) \end{aligned}$$

$$\Rightarrow LHS = RHS \neq$$

(b)

$$H(X) = -(p \ln p + (1-p) \ln (1-p))$$

$$\frac{d}{dp} H(X) = (\ln p + 1 - \ln (1-p) - 1) = -\left(\ln \frac{p}{1-p}\right) = \ln \frac{1-p}{p}$$

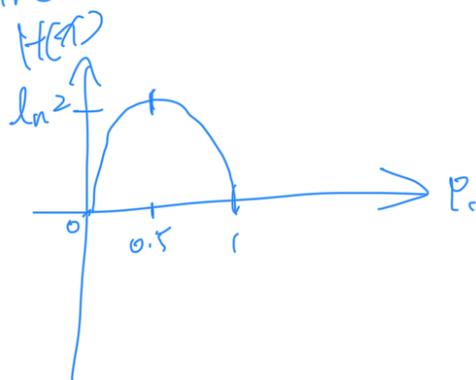
when $p = 0.5$, $H(X)$ is local max or min

$$\Rightarrow -(0.5 \ln 0.5 + 0.5 \ln 0.5) = \ln 2$$

when $p \rightarrow 1$ or $p \rightarrow 0$

$$\lim_{p \rightarrow 0} H(X) = 0 \text{ and } \lim_{p \rightarrow 1} H(X) = 0$$

$\Rightarrow$ so the Max of $H(X)$ is $\ln 2$, and the Min of $H(X)$ is 0



(c)

$$H(X) = -\sum_{i=1}^{\infty} p_i \ln p_i$$

$$= -(p_1 \ln p_1 + p_2 \ln p_2 + p_3 \ln p_3 + \dots + p_n \ln p_n)$$

according the hint

$$-(p_1 \ln p_1 + p_2 \ln p_2 + p_3 \ln p_3 + \dots + p_n \ln p_n) \leq -[(\ln p_1)^{p_1} \times (\ln p_2)^{p_2} + \dots + (\ln p_n)^{p_n}]^{\frac{1}{n}}$$

$$\text{若希望有最大值, 则 } p_1 \ln p_1 = p_2 \ln p_2 = \dots = p_n \ln p_n$$

$$\Rightarrow p_1 = p_2 = p_3 = \dots = p_n = \frac{1}{n}$$

$$\therefore \text{Max of } H(X) = -\sum_{i=1}^n \frac{1}{n} \ln \left(\frac{1}{n} \right) = -\sum_{i=1}^n \frac{1}{n} \ln n = \ln n \neq$$

(d)

Min of $H(X) = 0$, which occur when $p_i = 0 \quad \forall i (1 \leq i \leq n; i \in N) \neq i$
 $p_j = 0$

#4

coy:

$$p(X=k) = (1-p)^{k-1} p, \quad k=1, 2, 3, \dots$$

$$E[X] = \sum_{k=1}^{\infty} k p(X=k) = \sum_{k=1}^{\infty} k (1-p)^{k-1} p = p \sum_{k=1}^{\infty} k (1-p)^{k-1} = p \sum_{k=1}^{\infty} \frac{1}{p^2} = \frac{1}{p}$$

(due to $\sum_{k=1}^{\infty} k x^{k-1} = \frac{1}{(1-x)^2}$)

$$\begin{aligned}
 i) \quad E[e^{tx}] &= \sum_{k=1}^{\infty} e^{tk} p(x=k) = \sum_{k=1}^{\infty} e^{tk} (1-p)^{k-1} p = p \sum_{k=1}^{\infty} e^{tk} (1-p)^{k-1} \\
 &= \frac{p}{1-p} \sum_{k=1}^{\infty} e^{tk} (1-p)^{k-1} = \frac{p}{1-p} \cdot \frac{e^t (1-p)}{1 - e^t (1-p)} = \frac{pe^t}{1 - e^t (1-p)} \#
 \end{aligned}$$

補正

$$t < \ln(1-p) \Rightarrow e^t < 1-p \Rightarrow (1-p)e^t < 1 \Rightarrow r < 1$$

cb)

$$\begin{aligned}
 \text{Var}[Z] &= E[Z^2] - (E[Z])^2 \\
 E[Z] &= \sum_{k=1}^{\infty} k p(Z=z_k) = \sum_{k=1}^{\infty} k \cdot \frac{6}{(\pi k)^2} = \sum_{k=1}^{\infty} \frac{6}{k \pi^2} \quad (k=1) \\
 &\Rightarrow E[Z^2] \text{ doesn't exist} \Rightarrow \text{Var}[Z] \text{ doesn't exist.}
 \end{aligned}$$

$$\sum_{n=1}^{\infty} Z_n^3 p_Z(Z_n) = \sum_{n=1}^{\infty} (-1)^n n \cdot \frac{6}{(\pi n)^2} = \sum_{n=1}^{\infty} \frac{(-1)^n 6}{n \pi^2} \doteq -0.368 \#$$

$E[Z^3]$ is said to exist if $E[|Z|^3]$ exist.

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n 6}{\pi^2 n} \right| = \sum_{n=1}^{\infty} \frac{6}{\pi^2 n} \quad \text{according to p-series } p=1$$

so $\sum_{n=1}^{\infty} \frac{6}{\pi^2 n}$ diverge $\Rightarrow E[Z^3]$ doesn't exist.

#5

(a) 47.45%

(b)

Training-Test Split	Prior (0.5, 0.5)	Prior (0.6, 0.4)	Prior (0.7, 0.3)	Prior (0.8, 0.2)
50%-50%	97.45%	97.95%	98.06%	98.35%
60%-40%	97.58%	97.85%	97.94%	98.21%
70%-30%	97.67%	97.67%	97.97%	98.09%
80%-20%	97.40%	97.58%	97.94%	98.21%

我使用了不同的訓練集/測試集劃分（50%-50%，60%-40%，70%-30%，80%-20%）以及不同的 prior distributions（0.5,0.5；0.6,0.4；0.7,0.3；0.8,0.2）來測試分類器對垃圾訊息的準確性。從表中的結果可以觀察到一些規律

prior distributions 的影響：

當 prior distributions 中“spam”佔比提高時（例如 0.8, 0.2），分類器的準確率有輕微提升。

最佳準確率出現在 50%-50%數據劃分且 prior distributions 為 (0.8, 0.2) 時，。這說明在更傾向認為消息為垃圾短信時，分類器能更好地識別垃圾訊息。

數據劃分的影響：

數據劃分對準確率的影響相對較小。在所有不同的數據劃分（50%-50%，60%-40%，70%-30%，80%-20%）中，準確率始終保持在 97% 以上。

雖然準確率隨著訓練集比例的增大有所不同，但差異不大，說明分類器在不同訓練集大小下能夠穩定表現。